

Vaani Bhatnagar

vaanibhatnagar016@gmail.com | (609) 975-2021 | [Github](#) | [LinkedIn](#) | [Devpost](#)

EDUCATION

Olin College of Engineering

Needham, MA | May 2025 | 3.83/4.0

Bachelor's of Science in Mechanical Engineering

SKILLS

Fabrication: SolidWorks, Onshape, AutoCAD, Tormach Mill, Lathe, 3D-Printing, Band Saw, Drill Press, Belt/Disk Sander, Soldering

Design: Figma, InDesign, Photoshop, Illustrator, Product Design

Cloud/Databases: Snowflake, Azure (Data Factory), Postgres/Citus

Programming: Python, MATLAB, SQL, ReactJS, HTML, CSS, LaTeX, Arduino IDE, Atlas .ti

PROJECTS

Mehndi Bot (Mechanical Design, SolidWorks, 3D Printing, Laser Cutting, Rapid Prototyping)

- A Mehndi applicator robot with an XY gantry system for accurate henna application on a user's hand. Seamlessly integrating mechanical, software, and electrical components.

Rocky Robot (Control Systems Design, Dynamics)

- A control system for balancing an inverted pendulum on a moving cart with a 32U4 Balancing Robot Kit.
- Achieved system stabilization and produced detailed graphs of wheel speeds, angles, and positions over 65 seconds.

Orrery (Mechanical and Sustainable Design, Glassworking, Gear Mechanisms, Material Selection)

- Simulates relative positions and motions of celestial bodies with accurate gear ratios for realistic planetary orbits.
- Using scrap and reusable materials, features hand-torched glass planets for aesthetic appeal and sustainable design.
- Designed for easy assembly and disassembly to facilitate iteration and user replication.

Braildle (3D Design and Fabrication, Arduino, Pytesseract (Python), Tkinter)

- A refreshable display that uses camera input to convert conventional text into Braille. Designed, manufactured, and tested in 24 hours. Won first place at MakeMIT with a team of 4 female engineers.

EXPERIENCE

UI/UX Designer | Olin Senior Capstone Project in Engineering | Microsoft NERD

September 2024 - May 2025

- Co-managed a project to design and prototype a consent framework for health data that empowers marginalized communities, while collaborating with multiple liaisons as the communications lead, integrating user feedback into iterative design cycles, and creating scalable user research frameworks and practices to make evidence-based design decisions.

Operations Intern - Accelerate Wind (Remote)

September 2024 - December 2024

- Conducted research and crafted grant proposals to secure funding for Accelerate Wind's innovative wind-solar turbine system for affordable energy in commercial buildings.
- Promoted the company by pitching at startup competitions and conferences, driving mainstream adoption and growth for the integrated renewable energy solution.

Software Engineering Intern - Optum (Basking Ridge, NJ)

June 2024 - August 2024

- Designed and implemented a CI/CD data pipeline with Apache Kafka, GitHub Actions, AWS S3, and Java in an Agile/Scrum environment, automating ETL tool testing to enhance system security and performance significantly.
- Achieved a \$0.5 million profit boost by optimizing the pipeline's efficiency and reliability, showcasing advanced technical and engineering problem-solving skills.

Mechanical Design Engineer - NASA Nationwide Eclipse Ballooning Project (Needham, MA)

December 2023 - May 2024

- Designed and manufactured robust enclosures for high-altitude balloon missions, leveraging advanced structural mechanics to endure extreme conditions such as altitudes over 70,000 ft, winds above 120 mph, and unpredictable landing scenarios.
- Conducted extensive real-world testing to validate enclosure designs, ensuring critical circuitry and cameras remained operational in the most demanding environments, showcasing expertise in mechanical durability and reliability.

Formula SAE Cockpit Engineer - Olin Electric Motorsports (Needham, MA)

December 2023 - October 2024

- Led the design and development of a durable dashboard system with integrated LCD, E-Stop, and electrical modules, ensuring ergonomic optimization and a secure, waterproof enclosure for low-voltage circuit boards.
- Refined cockpit design through extensive testing of an adaptable mock rig, incorporating user feedback to significantly enhance ergonomic efficiency and user experience of the cockpit.

Systems Lead Engineer - Public Interest Technology & Boston Food Forest Coalition (MA)

September 2023 - May 2024

- Designed a solar-powered drip irrigation system for the Boston Food Forest Coalition, applying fluid dynamics and thermodynamics to optimize efficiency and serviceability.
- Developed a customizable model using an algorithm to account for crops, elevation, rainfall, and energy levels, incorporating mechanical principles for adaptable irrigation solutions.

Data Science Intern - Optum (Basking Ridge, NJ)

June 2023 - August 2023

- Built a Snowflake Database that improved patient query speed by 100x and reduced costs by 79%, leveraging SQL, Python, and Azure Data Factory to automate data flow using Agile practices.
- Pitched advanced database designs to CXOs, highlighting progress in data migration and API development.

Product Design Intern - Optum (Basking Ridge, NJ)

June 2022 - August 2022

- Developed Go-To-Market strategies and Marketectures to unify product ecosystems and elevate customer experience.
- Led product research and competitor analysis to identify integration opportunities, presenting a streamlined multi-product model to the CTO that boosted governance and customer satisfaction.

IT Help Desk Technician - Olin IT Department (Needham, MA)

September 2021 - May 2025

- Expert in Dell laptop assembly, Windows management, dual-booting, imaging, and diagnosing BIOS, Linux, Ubuntu, Windows, and iOS issues.
- Independently complete work orders, assisting up to 10 clients per shift and averaging 4 work orders while improving efficiency through team coordination.

RESEARCH AND PUBLICATIONS

Education Research: Learning Journeys

December 2021 - Present

- **Meta-Analysis of Student-Researchers' Learning Journeys Through a Reflective Practice**
IEEE Frontiers in Education Conference (FIE) 2024
- **Conforming and Non-Conforming Motivations and Behaviors in Students' Holistic Learning Journeys**
IEEE Frontiers in Education Conference (FIE) 2023
- **Majority or Minority: The Impact of Students' Identities on Their Learning Journeys**
IEEE Frontiers in Education Conference (FIE) 2022

NASA Nationwide Ballooning Project

December 2023 - May 2024

- **Olin College and Boston College's High Altitude Balloon Flights During the 2023 Annular and 2024 Total Solar Eclipses**
Academic High Altitude Conference 2024

CLUBS AND EXTRACURRICULARS

Thermodynamics and Transport Phenomenon II Course Assistant

January 2025 - Present

Engineering Systems Analysis Course Assistant

January 2025 - Present

Academic Resource Co-Designer

September 2024 - Present

Olin Dance Ensemble Lead Choreographer

January 2023 - Present

Olin South Asian Student Association President

January 2022 - Present

Collaborative Design Course Assistant

January 2024 - May 2024

Olin First Year Introduction Course Assistant

September 2023 - December 2023

Thermodynamics and Transport Phenomenon I Course Assistant

September 2023 - December 2023